



For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only.

PRAVATAS 10 & 20 MG TABLETS (Pravastatin Sodium Tablets 10 mg & 20 mg)

Name and strength of active ingredient

Pravastatin Sodium Tablets 10 mg
Pravastatin Sodium Tablets 20 mg

Dosage form

Uncoated Tablet

Product Description

Pravastatin Sodium Tablets 10 mg:

Yellow colored, rounded rectangular shaped, biconvex, uncoated tablets debossed 'PDT' on one side and '10' on the other side.

Pravastatin Sodium Tablets 20 mg:

Yellow colored, rounded rectangular shaped, biconvex, uncoated tablets debossed 'PDT' on one side and '20' on the other side.

Pharmacodynamic properties

Pharmacotherapeutic group: Serum lipid reducing agents/cholesterol and triglyceride reducers/ HMG-CoA reductase inhibitors.

Pravastatin is a competitive inhibitor of 3-hydroxy –3- methylglutaryl-coenzyme A reductase, the enzyme catalyzing the early rate limiting step in cholesterol biosynthesis, and produces its lipid lowering effect in two ways. Firstly with the reversible and specific competitive inhibition of HMG-CoA reductase, it effects modest reduction in the synthesis of intracellular cholesterol. This results in an increase in the number of LDL-receptors on cell surfaces and enhanced receptor mediated catabolism and clearance of circulating LDL-cholesterol.

Secondly, pravastatin inhibits LDL-production by inhibiting the hepatic synthesis of VLDL-cholesterol, the LDL-cholesterol precursor.

In both healthy subjects and patients with hypercholesterolemia, pravastatin sodium lowers the following lipid values: total cholesterol, LDL-cholesterol, apolipoprotein B, VLDL-cholesterol and triglycerides; while HDL-cholesterol and apolipoprotein A are elevated.

Pharmacokinetic properties

Absorption:

Pravastatin is administered orally in the active form. It is rapidly absorbed; peak serum levels are achieved 1 to 1.5 hours after ingestion. On average, 34% of the orally administered dose is absorbed, with an absolute bioavailability of 17%. The presence of food in the gastrointestinal tract leads to a reduction in the bioavailability, but the cholesterol-lowering effect of pravastatin is the same whether taken with or without food.

After absorption, 66% of pravastatin undergoes a first pass extraction through the liver, which are the primary site of its action and the primary site of cholesterol synthesis and clearance of LDL-cholesterol. In vitro studies demonstrated that pravastatin is transported into hepatocytes and with substantially less intake in other cells. In view of this substantial first pass through the liver, plasma concentrations or pravastatin have only a limited value in predicting the lipid-lowering effect. The plasma concentrations are proportional to the doses administered.

Distribution:

About 50% of circulating pravastatin is bound to plasma proteins. The volume of distribution is about 0.51 l/kg. A small quantity of pravastatin passes into the human breast milk.

Metabolism and elimination:

Pravastatin is not significantly metabolized by cytochrome P450 nor does it appear to be a substrate or an inhibitor of p-glycoprotein but rather a substrate of other transport proteins.

Following oral administration, 20% of the initial dose is eliminated in the urine and 70% in the faeces. Plasma elimination half life of oral pravastatin is 1.5 to 2 hours. After intravenous administration, 47% of the dose is eliminated by renal excretion and 53% by biliary excretion and transformation. The major degraded product of pravastatin is the 3- α -hydroxy isomeric metabolite. This metabolite has one – tenth to one- fortieth the HMG-CoA reductase inhibitor activity of the parent compound. The systemic clearance of pravastatin is 0.81l/h/kg and the renal clearance is 0.38l/h/kg indicating tubular secretion.

Indication/Usage

Therapy with pravastatin should be considered a component of multiple risk factor intervention in those individuals at increased risk for atherosclerotic vascular disease due to hypercholesterolemia. Pravastatin should be used in addition to a diet restricted in saturated fat and cholesterol when the response to diet and other nonpharmacological measures alone has been inadequate.

Prevention of Coronary Heart Disease:

In hypercholesterolemic patients without clinically evident coronary heart disease, Pravastatin is indicated as an adjunct to diet to reduce the risk of fatal and non-fatal myocardial infarction, need for myocardial revascularization procedures, and to improve survival by reducing cardiovascular deaths.

Myocardial/Infarction:

In patients with previous myocardial infarction and average (normal) serum cholesterol levels, Pravastatin is indicated to reduce the risk of recurrent myocardial infarction, need for myocardial revascularization procedures, and to reduce the risk of stroke and transient ischemic attacks (TIAs).

Atherosclerotic Disease Progression and Clinical Cardiovascular Events:

In hypercholesterolemic patients with atherosclerotic cardiovascular disease, pravastatin is indicated as an adjunct to diet to slow the progressive course of atherosclerosis and reduce the incidence of clinical cardiovascular events.

Hypercholesterolemia:

Pravastatin is indicated for the reduction elevated total-cholesterol and LDL-cholesterol levels in patients with primary hypercholesterolemia.

Prior to initiating therapy with pravastatin, secondary causes for hypercholesterolemia (e.g., obesity, poorly controlled diabetes mellitus, hypothyroidism, nephritic syndrome, dysproteinemias, obstructive liver disease, other drug therapy, alcoholism) should be excluded, and a lipid profile performed to measure Total-C, HDL-C, and TG. For patients with triglycerides (TG) less than 400 mg/dL, LDL-C can be estimated using the following equation:
$$\text{LDL-C} = \text{Total-C} - \text{HDL-C} - 1/5 \text{ TG}$$

For TG levels >400 mg/dL, this equation is less accurate and LDL-C concentrations should be determined by ultracentrifugation.

Periodic lipid determinations should be performed and dosage adjusted according to the patient's response to therapy.

Recommended Dose

Prior to initiating Pravastatin, the patient should be placed on a standard cholesterol lowering diet which should be continued during treatment.

The recommended dose is 10 to 20 mg once daily at bedtime. If serum cholesterol is markedly elevated (e.g., Total-C greater than 300 mg/dL), dosage may be initiated at 40 mg per day. Pravastatin may be taken without regard to meals.

Since the maximal effect of a given dose is seen within four weeks, periodic lipid determinations should be performed at this time and dosage adjusted according to the patient's response to therapy and established treatment guidelines. The recommended dosage range is 10 to 40 mg administered once a day at bedtime. Pravastatin may be given in divided doses.

In patients taking cyclosporine, with or without other immunosuppressive drugs, concomitantly with pravastatin, therapy should be initiated with 10 mg per day and titration to higher doses should be performed with caution. Most patients treated with this combination received a maximum pravastatin dose of 20 mg/day.

Concomitant Therapy

Some patients may require combination therapy with one or more lipid lowering agents.

Pharmacokinetic interaction studies with pravastatin administered concurrently with nicotinic acid, probucol, and gemfibrozil (see Warning and precautions, PRECAUTIONS, Skeletal Muscle) did not demonstrate any alterations in the bioavailability of pravastatin.

The lipid-lowering effects of Pravastatin on total and LDL cholesterol are enhanced when combined with a bile acid-binding resin. When administering a bile acid-binding resin (e.g., cholestyramine, colestipol) and pravastatin, Pravastatin should be given either one hour or more before or at least four hours following the resin.

Dosage in Patients Taking Cyclosporine

In patients taking cyclosporine, with or without other immunosuppressive drugs, concomitantly with Pravastatin therapy should be initiated with 10 mg/day and titration to higher doses should be performed with caution. Most patients treated with this combination received a maximum pravastatin dose of 20 mg/day.

Route of Administration

PRAVATAS 10/20 tablets: Orally

Contraindication

Hypersensitivity to the active substance or to any of the excipients. Acute liver disease or unexplained, persistent elevations of serum transaminase elevation exceeding 3 times the upper limit of normal.

Warnings & Precautions

HMG-CoA reductase inhibitors have been associated with biochemical abnormalities of liver function. As with other lipid-lowering agents, including nonabsorbable bile acid binding resins, increases in liver enzymes to less than three times the upper limit of normal have occurred during therapy with Pravastatin. The significance of these changes, which usually appear during the first few months of treatment initiation, is not known. In the majority of patients treated with pravastatin in clinical trials, these increased values declined to pretreatment levels despite continuation of therapy at the same dose.

As with other lipid-lowering agents, liver function tests should be performed periodically.

Special attention should be given to patients who develop increased transaminase levels.

REG-PRAVATAS-10-20(MAL)PIL

Size : 140 x 210 (mm) Front

Colour : Pantone Black

Date : 29/11/18, 10/02/20, 25/01/22, 19/04/22, 02/07/22

Liver function tests should be repeated to confirm an elevation and subsequent monitored at more frequent intervals. If increases in alanine aminotransferase (ALT) and aspartate aminotransferase (AST) equal or exceed three times upper limit of normal and persist, therapy should be discontinued.

Caution should be exercised when pravastatin is administered to patients with a history of liver disease or heavy alcohol ingestion.

Skeletal Muscle: Myalgia, myopathy and rhabdomyolysis have been reported with the use of HMG-CoA reductase inhibitors. Uncomplicated myalgia has rarely been reported in pravastatin-treated patients with an incidence similar to placebo. Rhabdomyolysis with renal dysfunction secondary to myoglobinuria has also very rarely been reported with pravastatin. However, myopathy should be considered in any patients with diffuse myalgia, muscle tenderness or weakness, and/or marked elevation of CPK. Patients should be advised to report promptly unexplained muscle pain, tenderness or weakness.

Pravastatin therapy should be discontinued if markedly elevated CPK levels occur or myopathy is suspected or diagnosed.

The risk of myopathy during treatment with another HMG-CoA reductase inhibitor is increased with concurrent therapy with either fibrates, cyclosporine, erythromycin or niacin. The use of fibrates alone is occasionally associated with myopathy. The combined use of pravastatin and fibrates should be generally being avoided.

Homozygous Familial Hypercholesterolemia: Pravastatin has not been evaluated in patients with rare homozygous familial hypercholesterolemia. In this group of patients, it has been reported that HMG-CoA reductase inhibitors are less effective because the patients lack functional LDL receptors.

Skeletal Muscle Effects

The use of fibrates alone may occasionally be associated with myopathy. The benefit of further alterations in lipid levels by the combined use of [Product Name] with fibrates should be carefully weighed against the potential risks of this combination.

Cases of myopathy, including rhabdomyolysis, have been reported with pravastatin co-administered with colchicine, and caution should be exercised when prescribing pravastatin with colchicine.

Pravastatin must not be co-administered with systemic fusidic acid. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. The patient should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness. Pravastatin therapy may be re-introduced seven days after the last dose of fusidic acid.

Interaction with other medicinal products and other forms of interaction

Colestyramine / Colestipol: When pravastatin was administered one hour before or four hours after colestyramine or one hour before colestipol and a standard meal, there was no clinically significant decrease in bioavailability or therapeutic effect of pravastatin.

Concomitant administration resulted in an approximately 40 to 50% decrease in the mean AUC of pravastatin.

Cyclosporine: Some investigators have measured cyclosporine plasma levels in patients on pravastatin, and to date, these results indicate no clinically meaningful elevations in cyclosporine levels. In one single-dose study, pravastatin plasma levels were found to be increased in cardiac transplant patients receiving cyclosporine.

Warfarin: Bioavailability parameters at steady state for pravastatin were not altered following administration with warfarin. Pravastatin did not alter the plasma protein binding of warfarin. Chronic dosing of the two products did not produce any changes in the anticoagulant action of warfarin.

Antipyrine: Clearance of antipyrine by the cytochrome P450 system was unaltered by concomitant administration of pravastatin. Since pravastatin does not appear to induce hepatic drug-metabolizing enzymes, it is not expected that any significant interaction of pravastatin with other drugs (eg: phenytoin, quinidine) that are metabolized by the cytochrome P450 system will occur.

Other drugs: Unlike most other HMG-CoA reductase inhibitors, pravastatin is not significantly metabolized by cytochrome P450 3A4. Therefore, plasma levels of pravastatin in vivo are not elevated when cytochrome P450 3A4 is inhibited by agents such as itraconazole or diltiazem.

Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Concomitant Therapy with Other Lipid Metabolism Regulators: Based on post-marketing surveillance, gemfibrozil, fenofibrate, other fibrates and lipid lowering doses of niacin (nicotinic acid) may increase the risk of myopathy when given concomitantly with HMG-CoA reductase inhibitors, probably because they can produce myopathy when given alone. Therefore, combined drug therapy should be approached with caution.

Gemfibrozil and nicotinic acid: Gemfibrozil and nicotinic acid do not statistically significantly affect the bioavailability of pravastatin. However, in a limited size clinical trial, a trend toward CK elevations and musculoskeletal symptoms was seen in patients treated concurrently with pravastatin and gemfibrozil. Myopathy, including rhabdomyolysis, has occurred in patients who were receiving

coadministration of HMG-CoA reductase inhibitors with fibric acid derivatives and niacin, particularly in subjects with pre-existing renal insufficiency.

Cyclosporine: In a multicentre study, the AUC values of pravastatin were shown to be five-fold higher in the presence of cyclosporine. There was no accumulation of pravastatin after multiple doses

Clarithromycin, colchicine: The risk of myopathy/ rhabdomyolysis is increased with concomitant administration of clarithromycin or colchicine with pravastatin.

Fusidic acid: The risk of myopathy including rhabdomyolysis may be increased by the concomitant administration of pravastatin with systemic fusidic acid. Co-administration of this combination may cause increased plasma concentrations of both agents. The mechanism of this interaction (whether it is pharmacodynamics or pharmacokinetic, or both) is yet unknown. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination. If treatment with fusidic acid is necessary, pravastatin treatment should be discontinued throughout the duration of the fusidic acid treatment.

Pregnancy and Lactation

Pregnancy: Pravastatin is contraindicated during pregnancy and should be administered to women of childbearing potential only when such patients are unlikely to conceive and have been informed of the potential risk. Special caution is recommended in adolescent females of childbearing potential to ensure proper understanding of the potential risk associated with pravastatin therapy during pregnancy. If a patient plans to become pregnant or becomes pregnant the doctor has to be informed immediately and pravastatin should be discontinued because of the potential risk to the fetus.

Lactation: A small amount of pravastatin is excreted in human breast milk; therefore pravastatin is contraindicated during breastfeeding.

Adverse Reactions

Adverse effects reported with Pravastatin include:

- myalgia, myositis or rhabdomyolysis (fever, muscle aches or cramps; unusual tiredness or weakness)
- Constipation, diarrhea, gas, heartburn, or stomach pain, dizziness, headache, nausea and skin rash
- Impotence and insomnia

Effects on ability to drive and use machines

Pravastatin has no or negligible influence on the ability to drive and use machines. However, when driving vehicles or operating machines, it should be taken into account that dizziness may occur during treatment.

Signs & Symptoms of overdose and Treatment

The experience with overdose of Pravastatin sodium in humans is very limited. There is no specific treatment in the event of overdose. In the event of an overdose, the patient should be treated symptomatically and supportive measures instituted as required.

Package

Pravastatin Tablets are packed in Alu-Alu blisters and packs of 30 tablets (3x10).

Storage

Do not store above 30°C. Keep out of the reach of children.

Shelf life

36 months

Manufactured by:

Intas Pharmaceuticals Limited
Plot numbers 457, 458 & 191/218P,
Sarkhej-Bawla Highway,
Matoda, Sanand, Ahmedabad,
Gujarat, IN-382210, INDIA

Product Registration Holder

Jetpharma Sdn Bhd
No.13, Jalan Rajawali 2,
Bandar Puchong Jaya
47100 Puchong
Selangor, Malaysia

Date of Revision: Feb 10, 2020



REG-PRAVATAS-10-20(MAL)PIL

Size : 140 x 210 (mm) Back

Colour : Pantone Black

Date : 29/11/18, 10/02/20, 25/01/22, 19/04/22, 02/07/22